Contents lists available at ScienceDirect

Chemical Engineering Research and Design

journal homepage: www.elsevier.com/locate/cherd

Dual-enhanced steam methane reforming by membrane separation of H_2 and reactive sorption of CO_2

Xiang Wu[a], *Cheng Wu*[a], *Sufang Wu*[a,b,*]

[a] College of Chemical and Biological Engineering, Zhejiang University, Hangzhou 310027, China
[b] Key Laboratory of Biomass Chemical Engineering of Ministry of Education, Zhejiang University, Hangzhou 310027, China

ARTICLE INFO

Article history:
Received 14 May 2014
Received in revised form 7 February 2015
Accepted 15 February 2015
Available online 20 February 2015

Keywords:
Hydrogen
Carbon dioxide
Steam methane reforming
Separation
Sorption enhancement
Membrane

ABSTRACT

This study proposes a dual-enhanced steam methane reforming (SMR) method for hydrogen production with simultaneous H_2 membrane separation and CO_2 sorption. Thermodynamic calculations and experimental studies were carried out to study the effectiveness of the dual-enhanced. Theoretically, dual-enhanced SMR was found to decrease the reaction temperature to 500 °C with a CH_4 conversion of 86.2% under the H_2 separation ratio of 0.6 and CO_2 sorption ratio of 0.95, in contrast, the CH_4 conversion was 52.1% under the same conditions but without any enhancement, thus lowering energy consumption by using dual-enhanced steam reforming. A palladium membrane tube was prepared by electroless plating and used for membrane separation of H_2. A fixed-bed reactor packed with complex catalyst containing nano-CaO as CO_2 sorbent and NiO/Al_2O_3 as reforming catalyst was assembled. Experimental results indicated that with a lowered reaction temperature to 500 °C, the CH_4 conversion can be 62.6%. The conversion of CH_4 by dual-enhanced SMR was 27.2%, higher than that obtained by conventional SMR at a reaction temperature of 600 °C and a steam-to-methane (S/M) molar ratio of 4. Product gas with 98.1% H_2 was directly obtained by dual-enhanced SMR.
© 2015 The Institution of Chemical Engineers. Published by Elsevier B.V. All rights reserved.

1. Introduction

As a clean energy carrier, hydrogen is mainly used in proton exchange membrane fuel cell (PEMFC). Currently, steam methane reforming (SMR) is the most economic method for H_2 production (Harrison, 2008). However, the gas obtained by SMR contains nearly 8–10% (v/v) CO (Barelli et al., 2008), which is poisonous to the Pt electrode in PEMFC, that its concentration in H_2 should be reduced to low levels (ppm). Methods to convert or remove CO include chemical reactions such as water gas shift (WGS) reaction (Tabakova et al., 2007; Goscianska et al., 2010), CO preferential oxidation (PROX) (Mariño et al., 2008; Holladay et al., 2009), methanation reaction (Takenaka et al., 2004; Schlereth and Hinrichsen, 2014; Zhang and Xu, 2007), and physical separation methods such as pressure swing adsorption (Majlan et al., 2009; Ruthven et al., 1994; You et al., 2012) and membrane separation (Acha et al., 2012; Matsuka et al., 2013).

Membrane separation uses palladium membrane to separate H_2, to obtain high-purity H_2 (Hwang et al., 2012; Patil et al., 2006; Sarić et al., 2012; Chen and Lu, 2012). However, contamination by CO in H_2 produced from SMR remains a problem when employing palladium membrane reactor (Mori et al., 2008; Adris et al., 1994). Mori et al. (2008) filled the methanation catalyst, Ru/Al_2O_3, in the palladium membrane reactor to reduce CO. However, the noble Ru catalyst is expensive, which

* Corresponding author at: Zhejiang University, College of Chemical and Biological Engineering, No. 38 Zheda Road, Hangzhou 310027, Zhejiang, China. Tel.: +86 0571 87953138; fax: +86 0571 87953735.
E-mail address: wsf@zju.edu.cn (S. Wu).
http://dx.doi.org/10.1016/j.cherd.2015.02.010
0263-8762/© 2015 The Institution of Chemical Engineers. Published by Elsevier B.V. All rights reserved.

is not economic to be applied in the H_2 production process.

On the other hand, sorption-enhanced reforming process (SERP) was proposed to enhance the SMR reaction by removing CO_2 *in situ* (Hildenbrand et al., 2006; Xiu et al., 2003; Harrison and Han, 1994; Hufton et al., 1999). Recently, reactive sorption-enhanced reforming (ReSER) process involves the use of sorption complex catalyst to directly obtain high-purity H_2 (95.5% v/v) with low CO content (<1%) was developed (Sufang et al., 2010). The excellent performance of nano-CaO in the complex catalyst as CO_2 sorbent ensures that the sorption complex catalyst can be used efficiently in reforming of methane for H_2 production (Xie et al., 2012; Butler et al., 2011; Arstad et al., 2009; Solieman et al., 2009). There are few reports on combining the membrane separation with sorption enhancement to produce H_2. Marono et al. (2014) packed the CO_2 adsorbent in a Pd-membrane reactor to study the water gas shift (WGS) reaction. For steam methane reforming, Andrés et al. (2011) packed the mixed CO_2 adsorbent catalyst and Pd-membrane in a fluidized-bed reactor to study the dual-enhancement on SMR. However, the mix of adsorbent particles with catalyst particles can result into large heat and mass transfer resistance between reaction and sorption. Besides, the limestone applied in the research has low adsorption capacity, which results into short enhancement regime. In this paper, we proposed the use of complex catalyst that contains adsorbent and catalyst in one particle, can achieve the goal of real "*in-situ*" sorption of CO_2, and thus lower the heat and mass transfer resistance during reaction and sorption. Moreover, in this study, the complex catalyst directly contacted with the membrane to separate the H_2 *in-situ*, achieving the goal of dual-enhancement, and comparison was also analyzed between the thermodynamic and experimental results of the process.

For the energy efficiency of these technologies, such as sorption enhanced, membrane separation and dual-enhanced SMR, Tzanetis et al. (2012) used exergy analysis method to compare the sorption-enhanced SMR with conventional SMR, and found that a exergy decrease of 10.8% can be achieved by using sorption enhanced SMR. For membrane separation compared with conventional SMR, a H_2 purification can be eliminated, such as PSA, which makes the unit as compact as 1/3 in volume and 1/2 in terms of area of the conventional unit by applying membrane in SMR (Barelli et al., 2008), which reduce the overall cost of the plant. For dual-enhanced SMR, which combines the advantage of the sorption-enhancement and membrane-separation enhancement, can further increase the efficiency of the H_2 production by methane reforming.

In this paper, we proposed a method that combines reactive sorption complex catalyst with palladium membrane in one reactor to develop a dual-enhanced SMR system. Thermodynamic calculations and experimental investigations were conducted in detail to evaluate the effect of H_2 separation and/or CO_2 sorption on SMR and a comparison was made between thermodynamic calculation results and experimental results on dual-enhanced SMR.

2. Thermodynamic analysis model

The schematic of the experimental setup for dual-enhanced SMR is showed in Fig. 1. The fixed bed reactor packed the sorption complex catalyst on the outside of palladium membrane tube. The CO_2 produced by SMR was adsorbed by the sorbent, and H_2 was passed through the palladium tube to obtain high-purity H_2.

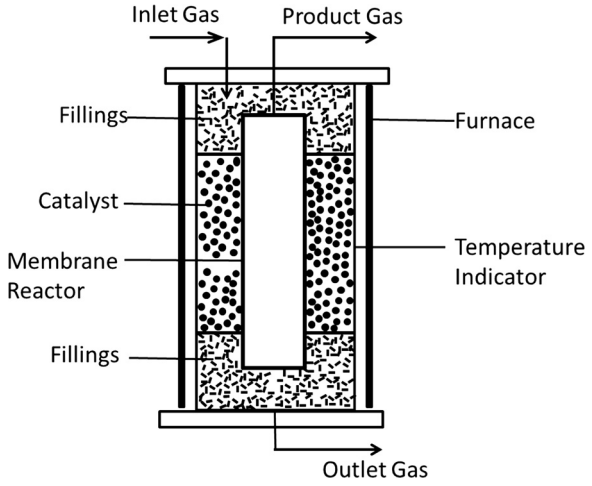


Fig. 1 – Illustration of dual-enhanced SMR reactor.

According to the reaction conditions of dual-enhanced SMR, the following assumptions were made to build the theoretical thermodynamic model:

(1) Equilibrium assumption and steady state operations;
(2) Uniform temperature and pressure distribution in the reactor;
(3) Ideal gas assumption;
(4) Only H_2 can go through the palladium membrane.

The process mainly carries out two reactions (Balasubramanian et al., 1999): SMR (reaction (1)) and WGS (reaction (2)).

$$CH_4 + H_2O \leftrightarrow CO + 3H_2,\ \Delta H^{\ominus}_{298K} = 206.2\ \text{kJ/mol} \quad (1)$$

$$CO + H_2O \leftrightarrow CO_2 + H_2,\ \Delta H^{\ominus}_{298\,K} = 241.1\,\text{kJ/mol} \quad (2)$$

The equilibrium constants, K_{as} and K_{aw} for reactions (1) and (2), respectively, can be expressed as follows (P stands for the total pressure of the reaction system):

$$K_{as} = P^2 \frac{[H_2]^3\,[CO]}{[CH_4]\,[H_2O]} \quad (3)$$

$$K_{aw} = \frac{[H_2]\,[CO_2]}{[CO]\,[H_2O]} \quad (4)$$

We define CO_2 sorption ratio as ζ to determine the percentage of CO_2 adsorbed by the sorbent, H_2 separation ratio as γ to determine the percentage of H_2 separated by the membrane, the conversion of the SMR reaction as X_S to evaluate the reaction extent of the SMR and numerically equals to the conversion of CH_4, the conversion of WGS reaction as X_W to evaluate the reaction extent of the WGS and numerically equals to the production rate of CO_2, the molar ratio of S/M as β. We can get the partial pressure of each component as follows:

$$P_{CH_2} = P\frac{1 - X_S}{1 + \beta + (2 - 3\gamma)\,X_S - X_S X_W\,(\gamma + \zeta)} \quad (5)$$

$$P_{H_2O} = P\frac{\beta - X_S - X_S X_W}{1 + \beta + (2 - 3\gamma)\,X_S - X_S X_W\,(\gamma + \zeta)} \quad (6)$$

Table 1 – Chemical composition of plating solution and the reaction conditions.

Component	Value
$PdCl_2$	2 g/L
Na_2EDTA	40 g/L
$NH_3{\cdot}H_2O$	120 mL/L
NaH_2PO_2	6 g/L
T	50 °C
pH	9–11

$$P_{CO_2} = P\frac{X_S X_W (1-\zeta)}{1+\beta+(2-3\gamma)X_S - X_S X_W(\gamma+\zeta)} \quad (7)$$

$$P_{H_2} = P\frac{X_S (X_W+3)(1-\gamma)}{1+\beta+(2-3\gamma)X_S - X_S X_W(\gamma+\zeta)} \quad (8)$$

$$P_{CO} = P\frac{X_S (1-X_W)}{1+\beta+2X_S - X_S X_W(\gamma+\zeta)+3\gamma X_S} \quad (9)$$

Eqs. (10) and (11) for the equilibrium constants are obtained by combining Eqs. (3)–(9).

$$K_{aw} = P^2\frac{X_S^4(X_W+3)^3(1-\gamma)^3(1-X_W)}{(1-X_S)(\beta-X_S-X_S X_W)\left[1+\beta+(2-3\gamma)X_S - X_S X_W(\gamma+\zeta)^2\right]} \quad (10)$$

$$K_{as} = \frac{X_S^2 (X_W+3)(1-\gamma)(1-X_W)}{X_S X_W (1-\zeta)(\beta-X_S-X_S X_W)} \quad (11)$$

These nonlinear equations are solved using Matlab. The conversions of each component on the dry basis are expressed according to Eqs. (12)–(15):

$$X_{CH_4}(\%) = \frac{1-X_S}{1-X_S X_W\zeta + X_S(X_W+3)(1-\gamma)} \times 100\% \quad (12)$$

$$X_{CO}(\%) = \frac{X_S(1-X_W)}{1-X_S X_W\zeta + X_S(X_W+3)(1-\gamma)} \times 100\% \quad (13)$$

$$X_{CO_2}(\%) = \frac{X_S X_W(1-\zeta)}{1-X_S X_W\zeta + X_S(X_W+3)(1-\gamma)} \times 100\% \quad (14)$$

$$X_{H_2}(\%) = \frac{X_S(X_W+3)(1-\zeta)}{1-X_S X_W\zeta + X_S(X_W+3)(1-\gamma)} \times 100\% \quad (15)$$

3. Experimental

3.1. *Preparation and characterization of palladium membrane*

Porous ceramic tube (provided by USTC, University of Science and Technology of China) with an average pore diameter of 0.2 μm was used as membrane support. The tube has an external diameter of 12 mm and an internal diameter of 8 mm. Electroless plating was used to deposit palladium on the tube (Zhang et al., 2008). The entire deposition process can be seen as referred to Gallucci's work (Gallucci et al., 2007). Table 1 lists the composition of the solutions used to prepare the palladium membrane. Scanning electron microscope (SEM, S-4800, HITACHI, Japan) was used to measure the thickness of the membrane.

Table 2 – Calculation conditions.

Cases	ζ	γ
1	0	0
2	0	0.3
3	0	0.6
4	0	0.9
5	0.95	0
6	0.95	0.3
7	0.95	0.6
8	0.95	0.9

3.2. *Scheme of ReSER process in palladium membrane reactor*

Fig. 2 shows a schematic diagram of the reaction system assembled for dual-enhanced SMR. A palladium membrane tube prepared in Section 3.1 was installed in the fixed bed reactor with an inner diameter of 3.3 cm. 5 g nano-CaO–NiO/Al_2O_3 (35.9 wt% CaO, 64.1 wt% NiO/Al_2O_3) complex catalyst with particle size of 0.8–1.3 mm was filled in the shell. The complex catalysis has a CO_2 sorption capacity of 1.6 mol/kg measured by Thermal-gravimetric Analysis (TGA, Pyris 1, PerkinElmer) under the temperature of 600 °C with a CO_2 partial pressure of 0.2 atm. The catalyst was prepared according to published literature (Sufang et al., 2010). Mass flow meters and gas concentration detectors were used to measure the flow rate and concentration of gas through the palladium membrane and the outlet, respectively. CH_4 conversion was defined by the following equation:

$$X_{CH_4}(\%) = \frac{F^0_{CH_4} - F_{out}\times y_{CH_4} - F'_{out}\times y'_{CH_4}}{F^0_{CH_4}} \times 100\% \quad (16)$$

where X_{CH_4} (%) is the conversion of CH_4; $F^0_{CH_4}$ is the feed rate of CH_4; F_{out} is the flow rate of gas through membrane; y_{CH_4} is the CH_4 content of gas through membrane; F'_{out} is the flow rate of the outlet product gas. y'_{CH_4} is the CH_4 content of the outlet gas.

4. Results and discussion

4.1. *Theoretical simulations to evaluate the effect of temperature*

Theoretical conditions selected to analyze the effect of CO_2 sorption and H_2 separation on CH_4 conversion is shown in Table 2. SMR is an endothermic reaction, and the temperature has a significant impact on the reaction. Higher temperature increases CH_4 conversion, but leads to more energy consumption. With this in view, a temperature range of 450 °C to 600 °C was selected to study the effect of temperature on CH_4 conversion and WGS reaction.

Fig. 3(a) and (b) shows the effect of reaction temperature and H_2 separation on CH_4 conversion without ($\zeta = 0$) and with ($\zeta = 0.95$) CO_2 sorption, respectively. Considering CH_4 conversion, it is evident that simultaneous separation of H_2 and CO_2 gives higher conversion than those separate only H_2 or only CO_2. With a reaction temperature of 600 °C, pressure of 1.3 atm, and S/M = 4, when $\zeta = 0.95$ and $\gamma = 0.3$ (Fig. 3(b)), the conversion of CH_4 is 98.0%, higher than 85.9%, which is the CH_4 conversion without separation of H_2 and CO_2 ($\zeta = 0$; $\gamma = 0$).

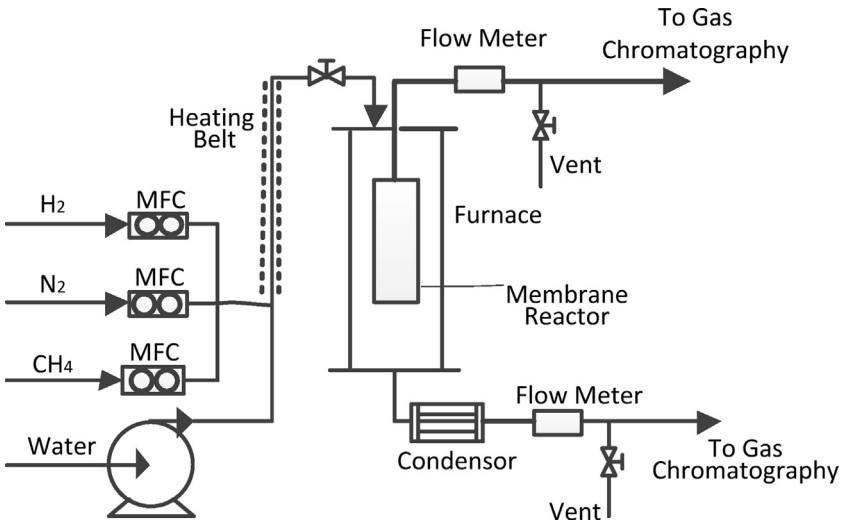


Fig. 2 – Process diagram of SMR in palladium membrane reactor.

To obtain a 98% CH_4 conversion but without enhancement, the temperature would be >650 °C theoretically under the same S/M and pressure. So, dual-enhanced SMR with the CO_2 sorption and H_2 separation is able to decrease the reaction temperature by >50 °C. Based on the above analysis and a published literature (Sufang et al., 2010), we used 600 °C as the reaction temperature for further thermodynamic calculations.

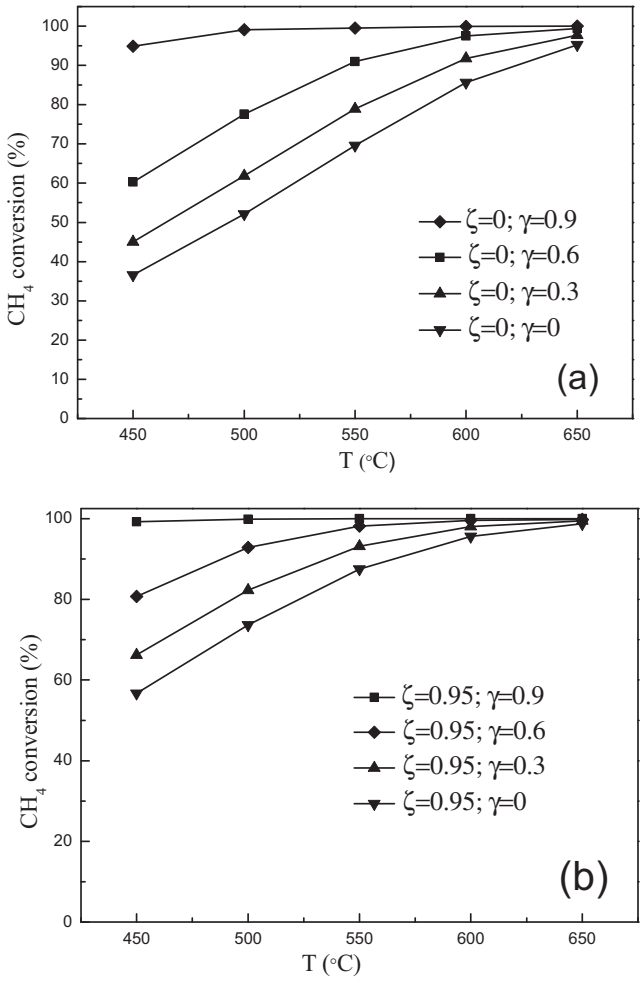


Fig. 3 – Influence of temperature on methane conversion (a) in the absence of CO_2 adsorption ($\zeta = 0$) and in the presence of CO_2 adsorption ($\zeta = 0$) under the reaction condition of $P = 0.1$ MPa, S/C = 4.

4.2. *Effect of CO_2 sorption and H_2 separation on methane conversion*

The effect of CO_2 sorption ratio (ζ) and H_2 separation ratio (γ) on the CH_4 conversion and WGS reaction were calculated as showed in Fig. 4(a) and (b), respectively. With only the sorption of CO_2, the CH_4 conversion increases to 86.1% while the simultaneous separation of H_2 and sorption of CO_2 increases the conversion to nearly 98% (Fig. 4(a)). It is obvious that with γ lower than 0.8, reforming conversion increases significantly irrespective of the value of ζ. However, when γ is larger than 0.8, the increase of reforming conversion is not assured for any ζ. In Fig. 4(b), increased γ leads to higher water-gas shift reaction conversion, the trend is more obvious for $\gamma < 0.6$. Considering the efficiency of palladium membrane towards H_2 separation and nano-CaO-based sorbent for CO_2 sorption, a choice combining the two materials ensures a high CO_2 sorption ratio under low H_2 separation ratio in the reforming reaction. Considering the effect of ζ and γ on WGS conversion, it is clear to see that irrespective of the value of CO_2 sorption ratio and H_2 separation ratio, the WGS reaction is enhanced.

4.3. *Effect of CO_2 sorption and H_2 separation on concentration of outlet gas*

Fig. 5(a)–(d) illustrates the effect of ζ and γ on the concentration of different components in the outlet gas. Fig. 5(a) shows the changes in CH_4 concentration with variations in ζ and γ. Increasing ζ effectively reduces the concentration of CH_4, which is consistent with the observed trend (Fig. 4(a)).

In Fig. 5(b), when only H_2 separation is carried out, CO concentration increases initially and then decreases, with a peak value at $\gamma = 0.7$, whereas the CO concentration decreases consistently with increase in CO_2 sorption ratio, ζ. Besides, the use of sorbent to adsorb CO_2 before membrane separation not only reduces CO content but also helps maintain the permeability and life extends of the palladium membrane.

It is obviously to see the effect of ζ and γ on CO_2 concentration in outlet gas by cursory examination of the reaction (Reactions (1) and (2), and Fig. 5(c)). An increase in γ cannot reduce CO_2 concentration, because separation of H_2 will

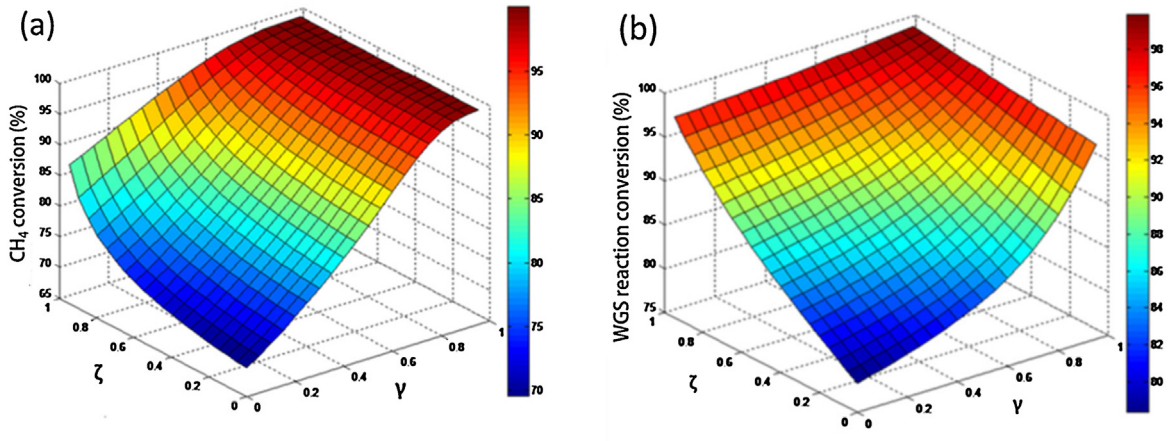


Fig. 4 – The effect of CO_2 adsorption ratio and H_2 separation ratios on (a) the conversion of methane and (b) water gas shift reaction conversion ($T = 550\,°C$, $P = 0.1\,MPa$, S/C = 4).

increase the CO_2 content in the reactor. Therefore, the use of the complex catalyst not only adsorbs the CO_2 to break the reforming reaction equilibrium but also increases the H_2 partial pressure, improving the H_2 permeation of membrane. Besides, the outlet gas can be reused or recycled when applied to industrial scale, the sorption of CO_2 can simplify its purification process of the outlet gas.

In Fig. 5(d), concentration of H_2 in the outlet gas increases (from 65% to 95%) with increase in γ, but decreases with increase in ζ, since more H_2 passes through the palladium membrane.

4.4. *Experimental study of dual-enhanced SMR*

Experimental studies were performed with the assembled reactor described in Sections 3.1 and 3.2 to compare with the theoretically determined results.

The thickness of palladium membrane was 7.2 μm, as showed in Fig. 6. To test whether the prepared palladium membrane can adequately be used in SMR, gas permeation experiments was performed to evaluate the membrane's selectivity and activity in the temperature ranging from 450 °C to 600 °C and under an outer-and-inner pressure difference

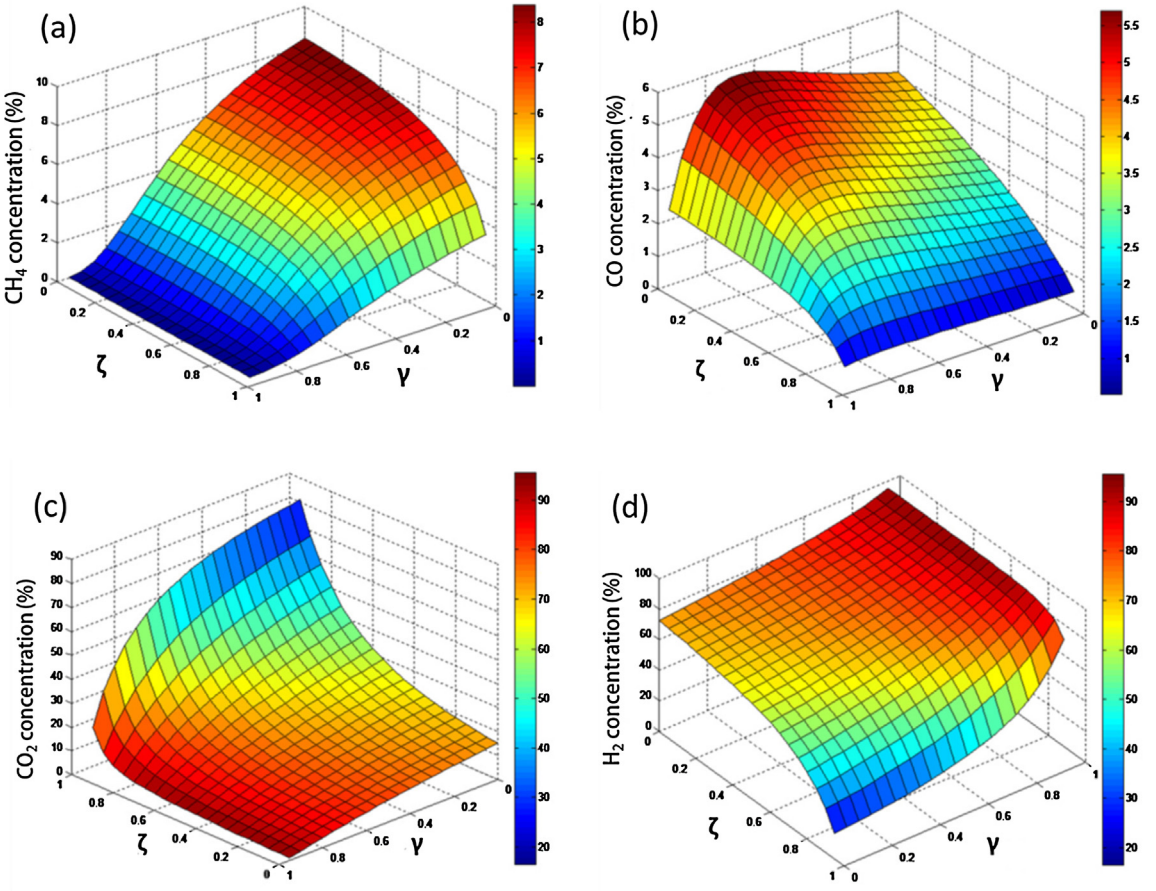


Fig. 5 – The effect of CO_2 adsorption ratio and H_2 separation ratio on (a) CH_4 concentration, (b) CO concentration, (c) CO_2 concentration, and (d) H_2 concentration in the exhaust gas. ($T = 550\,°C$, $P = 0.1\,MPa$, S/C = 4).

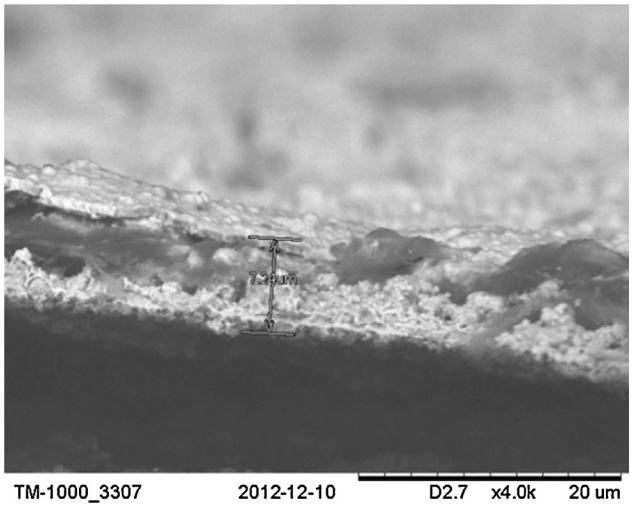


Fig. 6 – Cross-section SEM image of the palladium membrane.

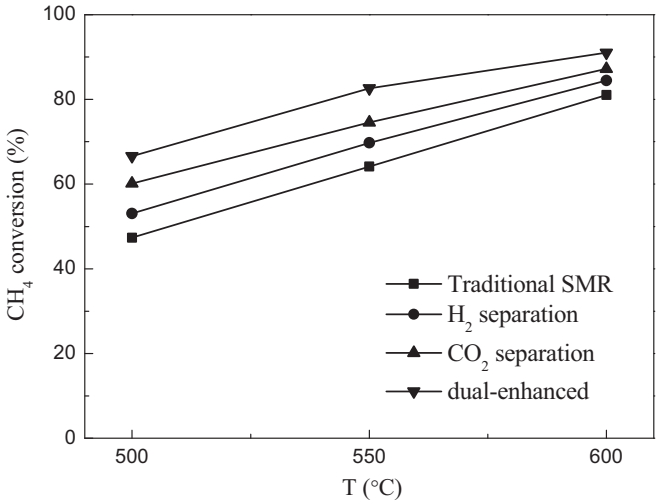


Fig. 8 – Effect of temperature on methane conversion under different enhancement methods.

of 30 kPa. The selectivity of the membrane was measured as the ratio of H_2 permeation flux towards N_2 flux under different temperatures and pressures, while the activity was measured as the H_2 permeation flux at different temperatures and pressures. A mix of H_2 and N_2 was used as test gas. The results are shown in Fig. 7. As the temperature increases from 450 °C to 600 °C, the permeation selectivity of the membrane for H_2 over N_2 is 9-fold higher. The membrane permeation flux for H_2 is described satisfactorily by Sieverts' law with activation energy, E_a, of 17.42 kJ/mol. The H_2 permeation flux for the palladium membrane was measured to be 0.015 mol/m^2/s at 600 °C and 1.3 atm. Compared with the works reviewed by Gallucci et al. (2013), especially for Shi's work (Shi et al., 2011), the membrane prepared and applied in this work is not good as the concerning membrane's activity and selectivity. We think this difference may due to the membrane type and preparation process. But the membrane's selectivity and activity prepared and applied in this study is still high, that can be adequately used in a dual-enhanced SMR process for H_2 separation.

Fig. 8 shows the CH_4 conversion under different conditions. CH_4 conversion is 91.0% with the sorption of CO_2 and membrane separation of H_2 at a reaction temperature of 600 °C, 1.3 atm, and S/M = 4. Even with a reduced reaction temperature of 550 °C, the CH_4 conversion (82.6%) is higher than that of the ordinary SMR reaction conversion at 700 °C with 3.2 MPa reaction pressure (80%). It appears that the sorption of CO_2 leads to a greater improvement on CH_4 conversion than that due to membrane separation of H_2. This can be due to near-complete sorption of CO_2 in the sorption-enhanced regime with nano-CaO as CO_2 sorbent or due to weak separation efficiency of the membrane in this dual-enhanced steam reforming reactor.

Fig. 9 shows the concentration of product gas through palladium membrane as function of time. A reaction enhanced regime of nearly 40 min was achieved because of the excellent performance of the nano-CaO sorbent and Pd-membrane used in the reaction system. With a flow rate of 80 mL/min to 90 mL/min, the concentration of H_2 in product gas is 98.1% and the CO content is <1000 ppm. The other components in the product gas are some minor CH_4 and CO_2. H_2 concentration declined with the reaction time extends, while CO_2 and CO concentrations increased with the breakthrough of sorbent, that the sorbent CaO converted to $CaCO_3$. For the outlet gas, CH_4 (~6.3%), CO_2 (~3.8%), and H_2 (~88%) with a flow rate of ~25 mL/min was obtained. The high content of H_2 in the outlet gas results from the high performance of the sorbent of CaO used in the complex catalyst, which adsorbs >95% of CO_2 from the gas phase. When the reaction temperature is lowered to 550 °C, the CH_4 conversion reduces to 82.6% and a significant quantity of CH_4 is detected in the outlet gas.

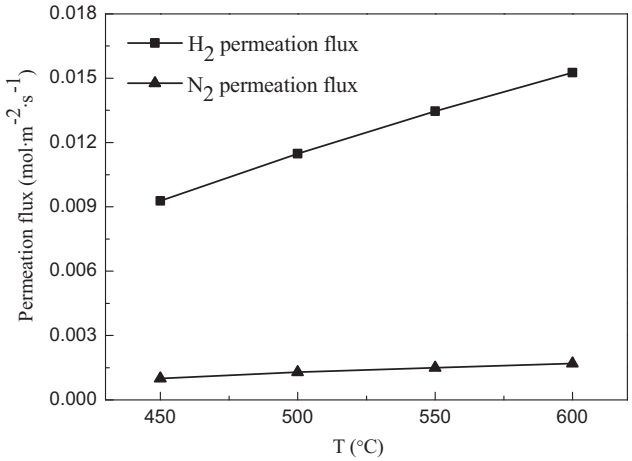


Fig. 7 – Temperature dependent hydrogen flux permeability (H_2: N_2 (v/v) = 9, test gas flow rate = 100 mL/min).

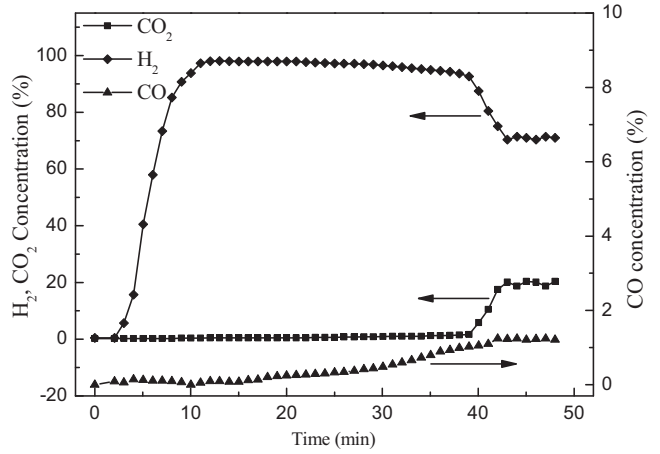


Fig. 9 – Concentration of gas as function of reaction time.

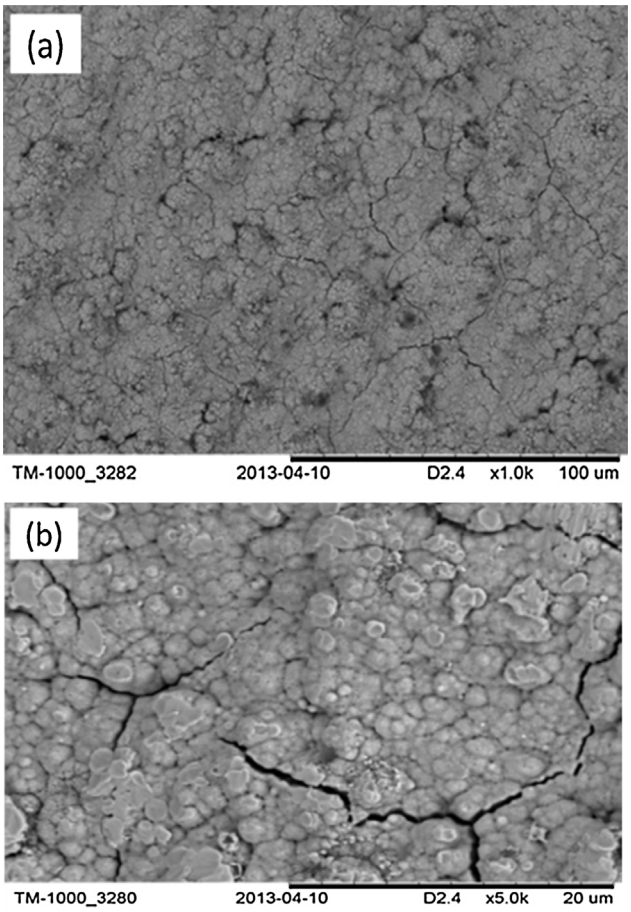


Fig. 10 – SEM image of palladium membrane after ten cycles of reaction, with amplification of 1000 (a) and 5000 (b).

Considering the gas concentration and flow rate in the product and outlet gases from the experimental result, it could found that a CO_2 sorption ratio of 0.96 and H_2 separation ratio of 0.8 has been achieved. Under the same theoretical condition, the CH_4 conversion would achieve 99% as the equilibrium conversion. However, because of the 100% selectivity of the membrane has not been fulfilled, the product gas still contains minor CH_4 and CO. Compared with the conventional SMR method, a decrease temperature from 800 °C to 600 °C and H_2 purity from 60 to 70% to more than 95% has been achieved. Compared with CO_2 sorption-enhanced method, a H_2 purity from 92% (Feng et al., 2012) to 98.1% has been achieved because of the *in situ* separation of H_2 also enhanced the reaction and improve the H_2 purity. However, it is obvious to see that by using high sorbent-to-catalyst ratio, high pressure difference in the reactor, and higher activity and selectivity of the membrane would result into better experimental results as higher H_2 purity in product gas and CH_4 conversion. From the above comparison among experimental results, thermodynamic calculation results, CO_2 sorption-enhanced results and ordinary SMR, the dual-enhanced SMR is a promising technology concerning lower reaction temperature with higher CH_4 conversion and H_2 purity.

However, about ten cycles of dual-enhanced steam methane reforming, the separation ability of membrane degraded. The SEM images of the membrane after ten reaction cycles are showed in Fig. 10. Many dark spots are shown in the membrane in Fig. 10(a) and those dark spots may be the deposited carbon, which may make the membrane gradually lose its separation ability. Meanwhile, many cracks are also seen in the membrane from Fig. 10(b).

As a difference with a ceramic support used in this study, Tanaka (Okazaki et al., 2008) reported the studies of using Al_2O_3 as membrane support. For Al_2O_3 supported membrane, H_2 permeability decreases when the temperature higher than 650 °C, and a strong interaction between Pd and Al was observed that may attribute to the decrease of the H_2 permeability. The results of Fig. 10(a) and (b) showed the ceramic supported membrane prepared and used in this study has a poor thermal stability. This may attribute to the mismatch of thermal expansion coefficient between the support and membrane, thus cause mechanical stress that leads to the cracks (Pacheco Tanaka et al., 2005).

5. Conclusions

This paper stimulates the dual-enhanced SMR, which involves simultaneous sorption of CO_2 and membrane separation of H_2.

Theoretical calculations show that independent separation of H_2 improves the CH_4 conversion to a larger extent than solely sorption of CO_2, but with the disadvantage of increased CO and CO_2 concentrations. Dual-enhanced SMR with the sorption and membrane separation not only improves the CH_4 conversion and get high-purity H_2, but also lower the reaction temperature and CO, CO_2 content. Experimental results indicate that the concentration of H_2 through the palladium composite membrane is 98.1% with the CO concentration <1000 ppm and a CH_4 conversion of 91.2% under 600 °C, reaction pressure of 1.3 atm, and S/M = 4. When the temperature is lowered to 550 °C, the CH_4 conversion of 82.6% from dual-enhanced SMR is still much higher than 64%, which is the theoretical value for CH_4 conversion obtained by conventional SMR under identical reaction conditions. The dual-enhanced has better performance than the CO_2 sorption-enhanced since the membrane separation also enhances the reaction to obtain high purity H_2 directly. From the perspective of reaction temperature, CH_4 conversion and H_2 purity, the dual-enhanced method is a promising technique to intensify the SMR to obtain high purity H_2 directly.

Acknowledgment

We gratefully acknowledge financial supports from the National Natural Science Foundation of China (NSFC, grant no. 21276234).

References

Acha, E., Requies, J., Barrio, V.L., et al., 2012. PdCu membrane applied to hydrogen production from methane. J. Membr. Sci. 415, 66–74.

Adris, A.M., Lim, C.J., Grace, J.R., 1994. The fluidized bed membrane reactor system: a pilot scale experimental study. Chem. Eng. Sci. 49, 5833–5843.

Andrés, M.B., Boyd, T., Grace, J.R., et al., 2011. In-situ CO_2 capture in a pilot-scale fluidized-bed membrane reformer for ultra-pure hydrogen production. Int. J. Hydrogen Energy 36, 4038–4055.

Arstad, B., Blom, R., Bakken, E., et al., 2009. Sorption-enhanced methane steam reforming in a circulating fluidized bed reactor system. Energy Procedia 1, 715–720.

Balasubramanian, B., et al., 1999. Hydrogen from methane in a single-step process. Chem. Eng. Sci. 15, 3543–3552.

Barelli, L., Bidini, G., Gallorini, F., et al., 2008. Hydrogen production through sorption-enhanced steam methane reforming and membrane technology: a review. Energy 33, 554–570.
Butler, James W.C., et al., 2011. CO_2 capture capacity of CaO in long series of pressure swing sorption cycles. Chem. Eng. Res. Des. 89, 1794–1804.
Chen, W.H., Lu, J.J., 2012. Hydrogen production from water gas shift reactions in association with separation using a palladium membrane tube. Int. J. Energy Res. 36, 346–354.
Pacheco Tanaka, David A., Llosa Tanco, Margot A., et al., 2005. Preparation of palladium and silver alloy membrane on a porous-alumina tube via simultaneous electroless plating. J. Membr. Sci. 247, 21–27.
Schlereth, David, Hinrichsen, Olaf, 2014. A fixed-bed reactor modeling study on the methanation of CO_2. Chem. Eng. Res. Des. 4, 702–712.
Majlan, E.H., Wan Daud, W.R., et al., 2009. Hydrogen purification using compact pressure swing adsorption system for fuel cell. Int. J. Hydrogen Energy 34, 2771–2777.
Feng, H.Z., Lan, P.Q., Wu, S.F., 2012. A study on the stability of a $NiO–CaO/Al_2O_3$ complex catalyst by La_2O_3 modification for hydrogen production. Int. J. Hydrogen Energy 37, 14161–14166.
Gallucci, F., Chiaravalloti, F., Tosti, S., et al., 2007. The effect of mixture gas on hydrogen permeation through a palladium membrane: experimental study and theoretical approach. Int. J. Hydrogen Energy 32, 1837–1845.
Gallucci, F., Fernandez, E., Corengia, P., et al., 2013. Recent advances on membranes and membrane reactors for hydrogen production. Chem. Eng. Sci. 92, 40–66.
Goscianska, J., Ziolek, M., Gibson, E., et al., 2010. Novel mesoporous zirconia-based catalysts for WGS reaction. Appl. Catal., B: Environ. 97, 49–56.
Harrison, D.P., 2008. Sorption-enhanced hydrogen production: a review. Ind. Eng. Chem. Res. 47, 6486–6501.
Harrison, D.P., Han, C., 1994. Simultaneous Shift reaction and carbon dioxide separation for the direct production of hydrogen. Chem. Eng. Sci. 49, 5875–5883.
Hildenbrand, N., Readman, J., Dahl, I.M., et al., 2006. Sorbent enhanced steam reforming (SESR) of methane using dolomite as internal carbon dioxide absorbent: limitations due to $Ca(OH)_2$ formation. Appl. Catal., A: Gen. 303, 131–137.
Holladay, J.D., Hu, J., King, D.L., et al., 2009. An overview of hydrogen production technologies. Catal. Today 139, 244–260.
Hufton, J.R., Mayorga, S., Sircar, S., 1999. Sorption-enhanced reaction process for hydrogen production. AIChE J. 45, 248–256.
Hwang, K.R., Lee, C.B., Ryi, S.K., et al., 2012. A multi-membrane reformer for the direct production of hydrogen via a steam-reforming reaction of methane. Int. J. Hydrogen Energy 37, 6601–6607.
Mariño, F., Baronetti, G., Laborde, M., et al., 2008. Optimized $CuO–CeO_2$ catalysts for CO-PROX reaction. Int. J. Hydrogen Energy 33, 1345–1353.
Marono, M., Barreiro, M.M., Torreiro, Y., Sánchez, J.M., 2014. Performance of a hybrid system sorbent–catalyst–membrane for CO_2 capture and H_2 production under pre-combustion operating conditions. Catal. Today 236, 77–85.
Matsuka, M., Higashi, M., Ishihara, T., 2013. Hydrogen production from methane using vanadium-based catalytic membrane reactors. Int. J. Hydrogen Energy 38, 6673–6680.
Mori, N., Nakamura, T., Sakai, O., et al., 2008. CO-free hydrogen production by membrane reactor equipped with CO methanator. Ind. Eng. Chem. Res 47, 1421–1426.
Okazaki, J., Ikeda, T., Pacheco Tanaka, D.A., et al., 2008. Strong interaction at the palladium/alumina interface of membrane during hydrogen permeation at elevated temperature. Chem. Lett. 9, 1004–1005.
Patil, C.S., et al., 2006. Experimental study of a membrane assisted fluidized bed reactor for H_2 production by steam reforming of CH_4. Chem. Eng. Res. Des. 5, 399–404.
Ruthven, D.M., Farooq, S., Knaebel, K.S., 1994. Pressure Swing Adsorption. VCH Publishers, New York, NY.
Sarić, M., van Delft, Y.C., Sumbharaju, R., et al., 2012. Steam reforming of methane in a bench-scale membrane reactor at realistic working conditions. Catal. Today 193, 74–80.
Shi, L., Goldbach, A., Xu, H., 2011. High-flux H_2 separation membranes from $(Pd/Au)_n$ nanolayers. Int. J. Hydrogen Energy 36, 2281–2284.
Solieman, A.A.A., Dijkstra, J.W., Haije, W.G., et al., 2009. Calcium oxide for CO_2 capture: operational window and efficiency penalty in sorption-enhanced steam methane reforming. Int. J. Greenhouse Gas Control 3, 393–400.
Sufang, W., Lianbao, L., Yanqing, Z., et al., 2010. A micro-sphere catalyst complex with nano $CaCO_3$ precursor for hydrogen production used in ReSER process. Eng. Sci. 8, 22–26.
Tabakova, T., Idakiev, V., Papavasiliou, J., et al., 2007. Effect of additives on the WGS activity of combustion synthesized CuO/CeO_2 catalysts. Catal. Commun. 8, 101–106.
Takenaka, S., Shimizu, T., Otsuka, K., 2004. Complete removal of carbon monoxide in hydrogen-rich gas stream through methanation over supported metal catalysts. Int. J. Hydrogen Energy 29, 1065–1073.
Tzanetis, K.F., Martavaltzi, C.S., Lemonidou, A.A., 2012. Comparative exergy analysis of sorption enhanced and conventional methane steam reforming. Int. J. Hydrogen Energy 37, 16308–16320.
Xie, M., Zhou, Z., Qi, Y., et al., 2012. Sorption-enhanced steam methane reforming by in situ CO_2 capture on a $CaO-Ca_9Al_6O_{18}$ sorbent. Chem. Eng. J. 207, 142–150.
Xiu, G., Li, P., Rodrigues, A.E., 2003. Adsorption-enhanced steam-methane reforming with intraparticle-diffusion limitations. Chem. Eng. J. 95, 83–93.
You, Y., Lee, D., Yoon, K., et al., 2012. H_2 PSA purifier for CO removal from hydrogen mixtures. Int. J. Hydrogen Energy 37, 18175–18186.
Zhang, X.L., Xiong, G.X., Yang, W.S., 2008. A modified electroless plating technique for thin dense palladium composite membranes with enhanced stability. J. Membr. Sci. 314, 226–237.
Zhang, Z., Xu, G., 2007. Thermally differential methanation—a novel method to realize highly selective removal of CO from H_2-rich reformates. Catal. Commun. 8, 1953–1956.